

Test Cases — Smart Scientific Calculator

1. Introduction

This document lists the test cases for the Smart Scientific Calculator, organised by iteration to match the incremental development model. Each case gives the steps, the expected result, the actual result observed and the pass/fail status. Supporting evidence is in the separate Test Evidence document.

2. Test Environment

Item	Detail
Application under test	Repository main branch (commit cff6fdc): public/index.html, public/script.js, server.js
Browser / UI	Google Chrome on macOS, http://localhost:3000
Runtime	Node.js v22.22.2, Express 4.22.3, Mongoose 8.x
Database	MongoDB (calculator_db); an in-memory stand-in was used for the automated run (see Test Evidence)
Execution	Test run of 3 October 2026: the real front-end and server code were driven by an automated script that clicks the actual buttons

3. Test Cases and Results

Iteration 1 - Basic Calculation

ID	Description	Steps	Expected Result	Actual Result	Status
TC-001	Addition of two positive integers	Click 5, +, 3, =	Display shows 8	Display shows 8	Pass
TC-002	Subtraction with a negative result	Click 2, -, 5, =	Display shows -3	Display shows -3	Pass
TC-003	Multiplication of decimals	Click 2, ., 5, ×, 2, ., 5, =	Display shows 6.25	Display shows 6.25	Pass
TC-004	Percentage calculation	Click 4, 5, %, =	Display shows 0.45	Display shows Error	Fail
TC-005	AC clears the display	Enter 123, click AC	Display resets to 0; history line cleared	Display shows 0; history line ""	Pass
TC-006	DEL removes the last character	Enter 123, click DEL	Display shows 12	Display shows 12	Pass

Iterations 2 & 3 - Scientific Functions

ID	Description	Steps	Expected Result	Actual Result	Status
TC-007	Square root of 144	Enter 144, click √	Display shows 12	Display shows 12	Pass
TC-008	Sine of 30 degrees	Enter 30, click sin	Display shows approximately 0.5	Display shows 0.49999999999999994	Pass
TC-009	Cosine of 60 degrees	Enter 60, click cos	Display shows approximately 0.5	Display shows 0.50000000000000001	Pass
TC-010	2 raised to the power 10	Enter 2, click x^y, enter 10 in the prompt, OK	Display shows 1024	Display shows 1024	Pass

Iteration 4 - Error Handling

ID	Description	Steps	Expected Result	Actual Result	Status
TC-011	Division by zero	Click 5, ÷, 0, =	Display shows Error	Display shows Error	Pass

ID	Description	Steps	Expected Result	Actual Result	Status
TC-012	Square root of a negative number	Click -, 9 (value -9), then √	Display shows Error	Display shows Error	Pass
TC-013	Malformed expression	Click +, +, =	Display shows Error	Display shows Error	Pass

Iteration 5 - Easy Interaction

ID	Description	Steps	Expected Result	Actual Result	Status
TC-014	Buttons give visual feedback on click	Click any button; inspect its press/hover style	Button shows press/hover feedback	Stylesheet defines .btn:active { transform: scale(0.95) } and a 0.2 s background-color transition	Pass
TC-015	Display updates in real time	Click 1, 2, 3 and observe the display	Display shows 1, then 12, then 123	Display shows 1, then 12, then 123	Pass

Iteration 6 - History & Database

ID	Description	Steps	Expected Result	Actual Result	Status
TC-016	Calculation is saved to the database	Click 7, +, 3, =; inspect POST /api/history	201 Created; expression "7 + 3", result "10"	201 Created; expression "7 + 3", result "10"	Pass
TC-017	History button shows recent calculations	With saved records present, click View Database History	Alert lists the 10 most recent calculations, newest first	Alert lists 10 records; first entry "1. 11 + 11 = 22"	Pass
TC-018	History button with an empty database	With no records, click View Database History	Alert "No history found in database."	Alert "No history found in database."	Pass
TC-019	History button when the server is down	Stop the server, click View Database History	Alert "Could not connect to database server."	Alert "Could not connect to database server."	Pass

4. Test Summary

Metric	Count
Total cases	19
Passed	18
Failed	1 (TC-004)

5. Defect Found

DEF-01 — % button does not compute a percentage (TC-004). Pressing 4, 5, %, = shows Error instead of 0.45. The % key appends the modulo operator to the expression, and the front end evaluates it with eval(), so "45 %" has no right-hand operand and raises a syntax error. The percentage() method in CalculationEngine is never called from the UI. The % key currently works as modulo (10 % 3 = 1, shown in the Test Evidence document). One possible fix is to make the % key append "/100" instead of the modulo operator; TC-004 should then be re-run.